



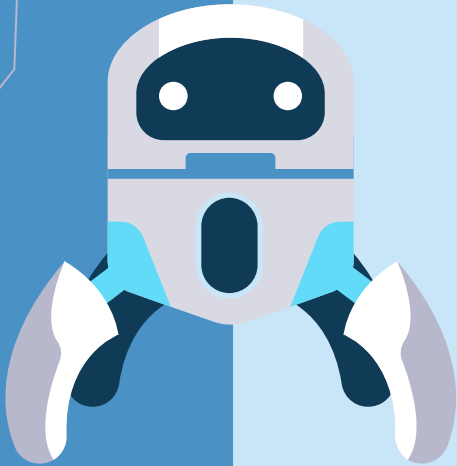
Wizardry Chess

Ricardo Metral, Max Eiken, Kevin Rivera



Project Idea

We want to create a fully autonomous chess board that can be played using just your voice. We want to mimic the idea of Wizardry Chess from Harry Potter. The pieces will also track a form of “loyalty” to a player based on the player’s rating during the game. “Insubordinate” pieces will purposefully disregard the players input and make their own decision.



■ Hardware Progress

- Electrical System

- Implemented all electrical components ✓
- Confirmed component functionality with Raspberry Pi ✓
- Powered everything from the 12V power supply ✓
- Made everything neat/hazard free...

- Gantry System

- Assembled gantry and tested functionality ✓
- Attached gears and pulleys to rollers and tested for resistance ✓
- Secured gantry system to table ✓
- 3D printed gantry attachment components (scrapped) ✗

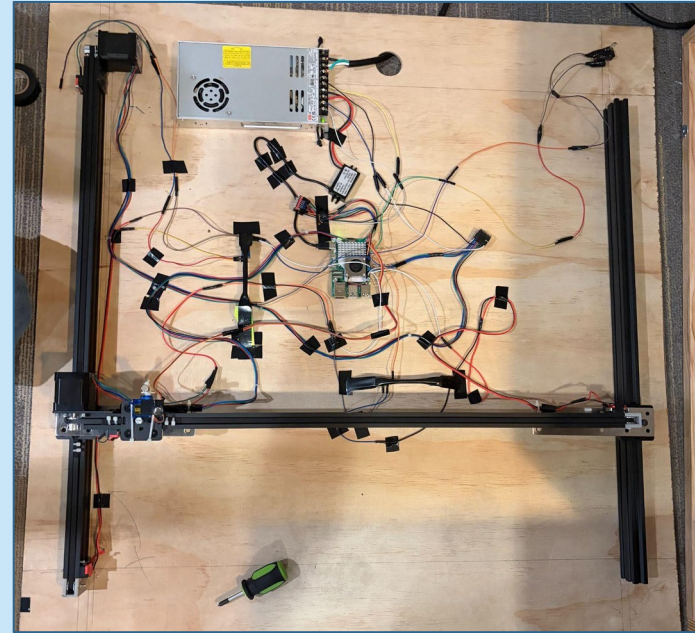
- Table

- The casing is complete ✓
- Outline sketch of the chessboard ✓
- Painting
- Legs ✓

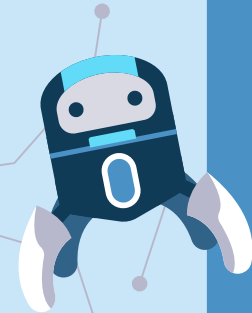
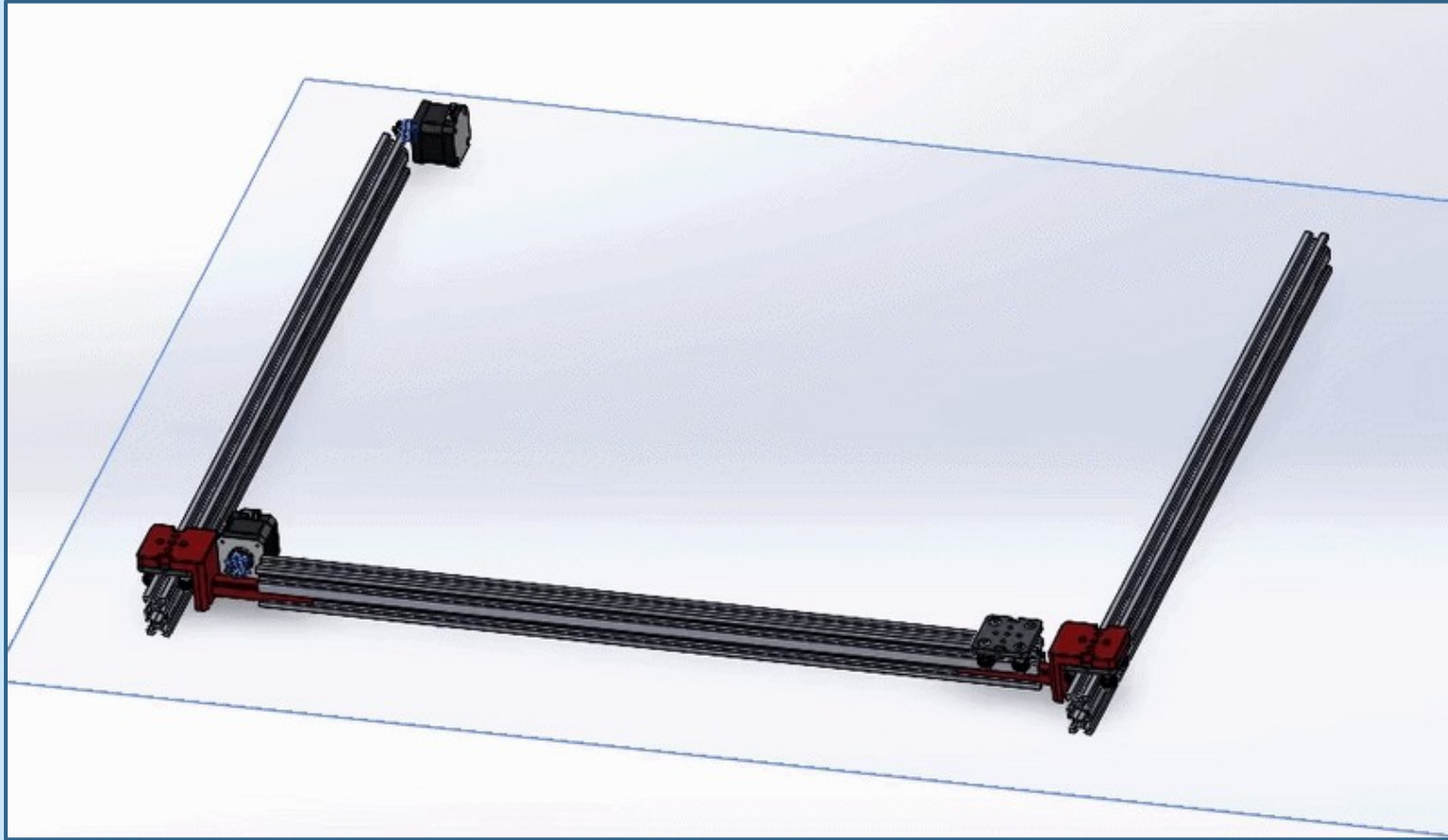
- Magnet

- Implemented electromagnet and MOSFET (scrapped) ✗
- Achieved magnet activation to grab and move pieces across board ✓
- Utilized permanent neodymium magnets actuated by servo to attract magnetic chess pieces ✓

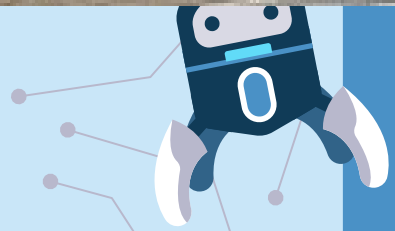
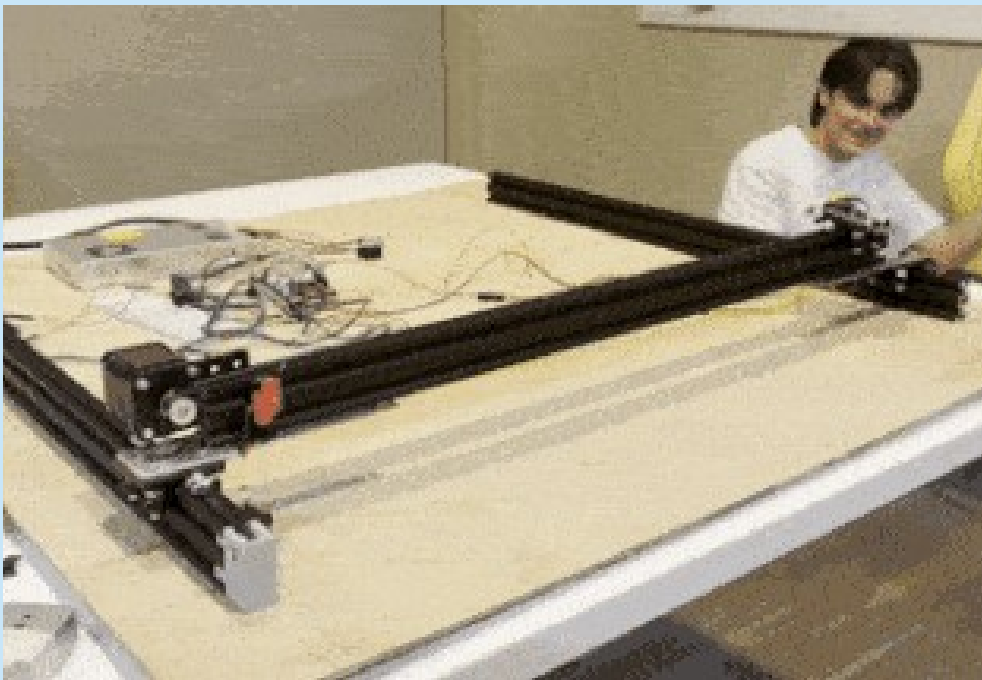
3x Stepper Motors	2x Servo Motors
1x Electromagnet	1x Raspberry Pi
3x 2040 Aluminum Extrusion	1x ESP32 Microcontroller
3x roller	1x Power Supply cord
3x Timing Belt	2x Stepper Mount Bracket
3x Pulleys	Wood
3x Idlers	3x Stepper Motor Drivers
6x Micro Limit Switch (DC)	1x MOSFET power controller
DC Power Supply	1x Microphone



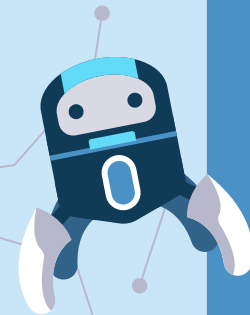
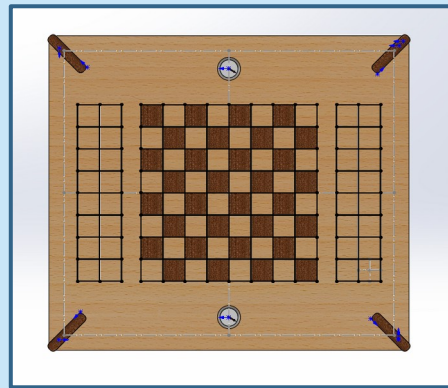
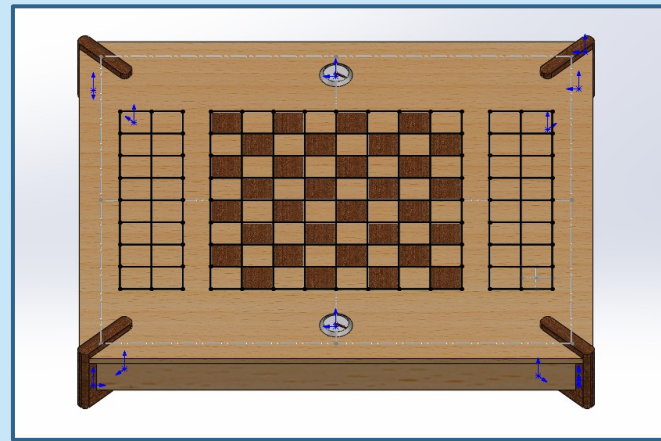
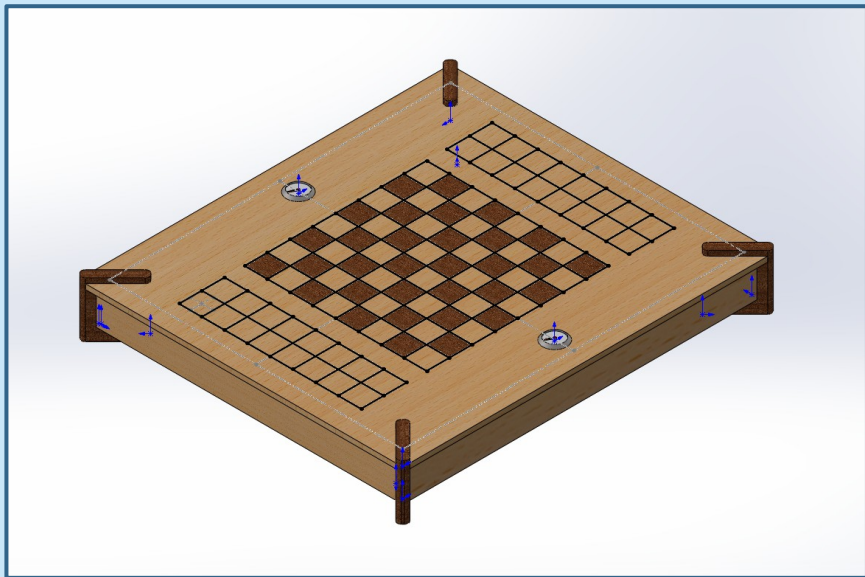
Prototype Gantry CAD



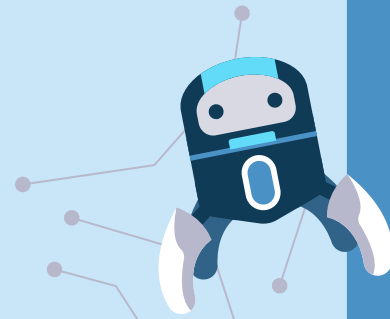
Updated Gantry Video



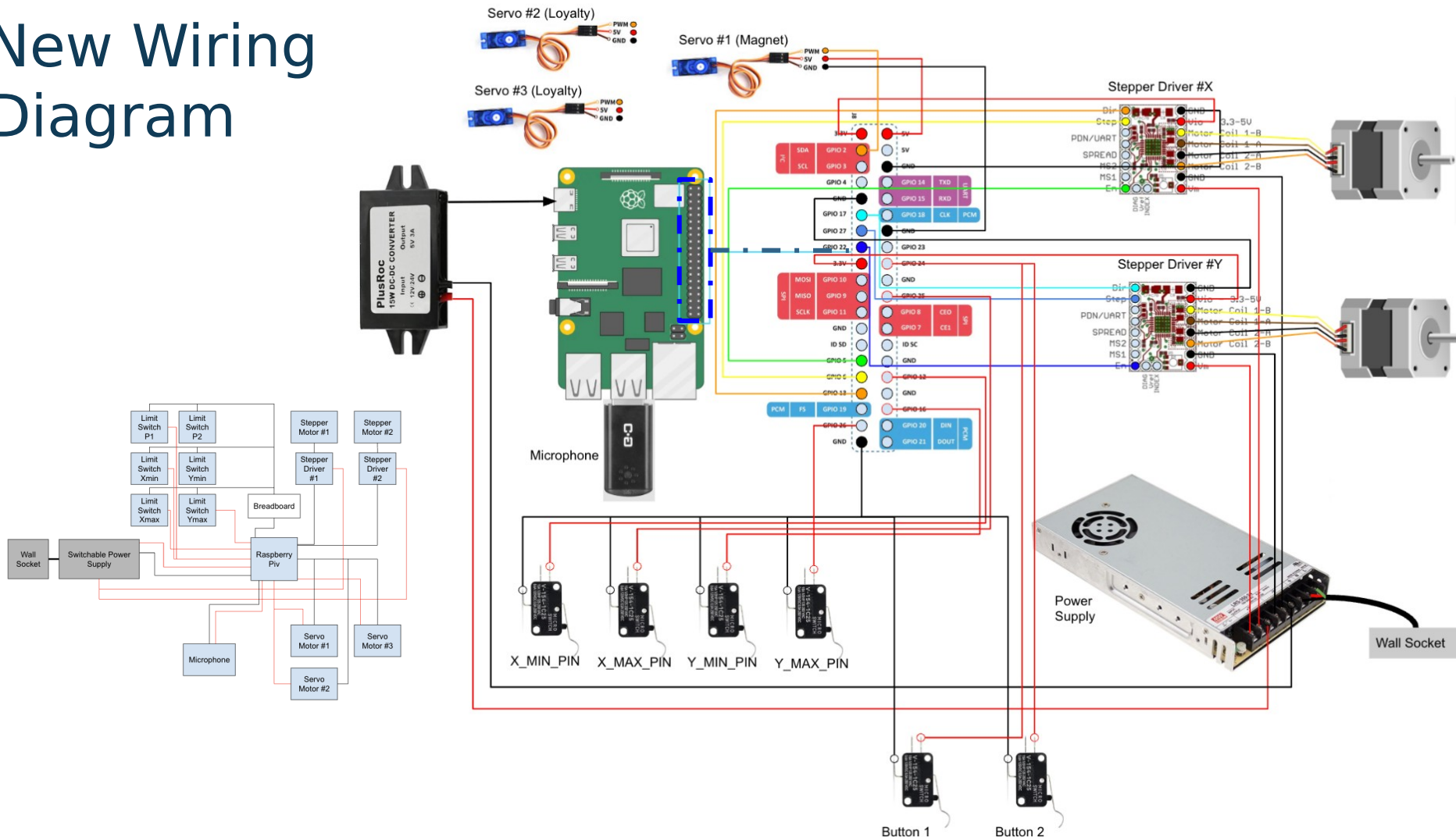
Prototype Table CAD



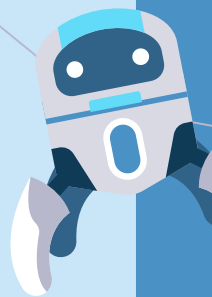
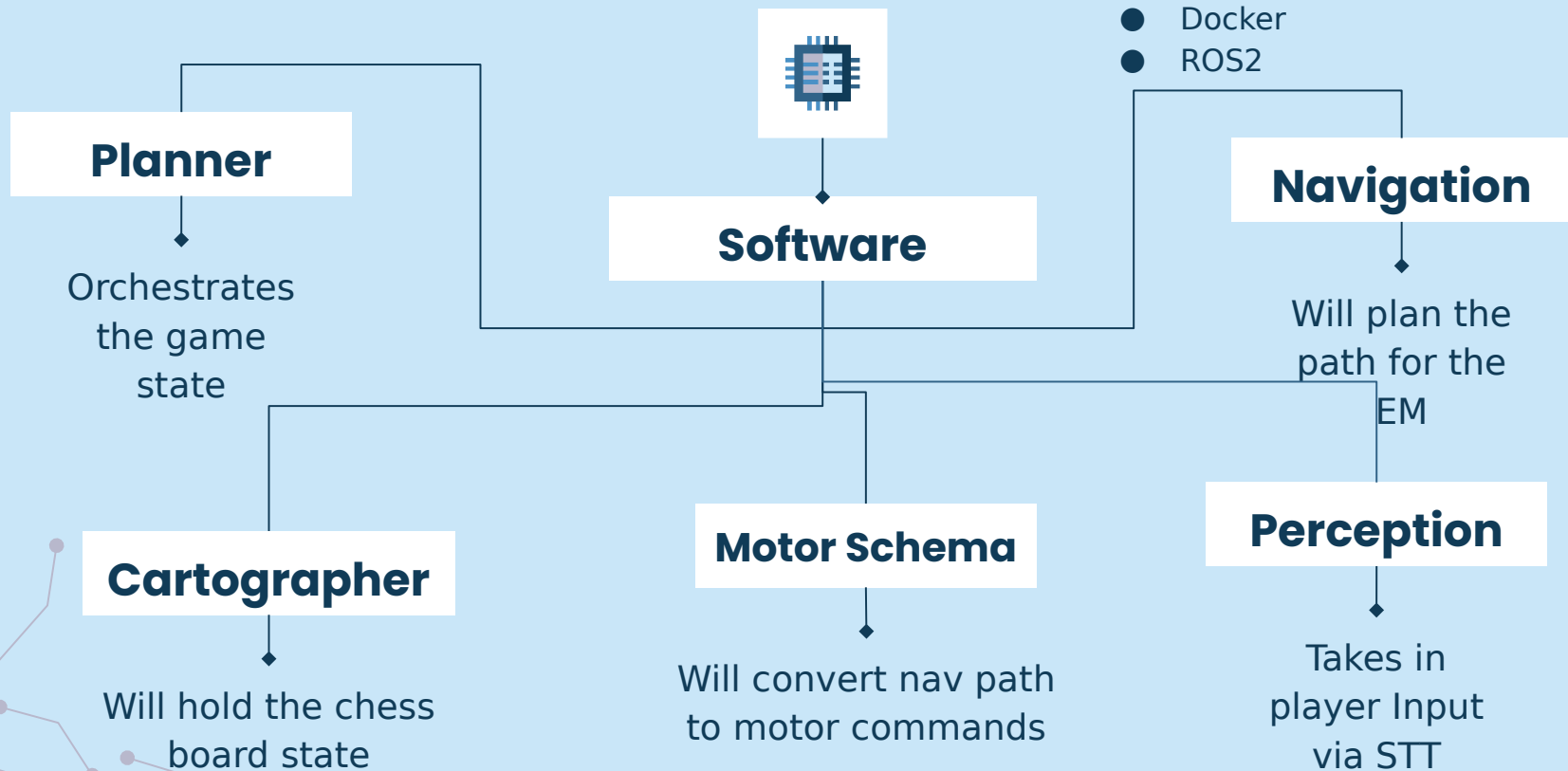
New Table Design



New Wiring Diagram



■ Software Subsystems



Software Progress

- Planning Subsystem:

- Receive input from player using perception subsystem topic ✓
- Publish player input to nav ✓
- Get loyalty system working ✓

- Perception Subsystem:

- Can receive player input via stt ✓
- Sends reset request ✓
- Can receive player input via grid clicking in Lichess ✓
- Blocks execution of next move until move is received ✓
- Sanitizes player input to make sure it is valid ✓
- Sends player move to planning subsystem ✓

- Cartographer Subsystem:

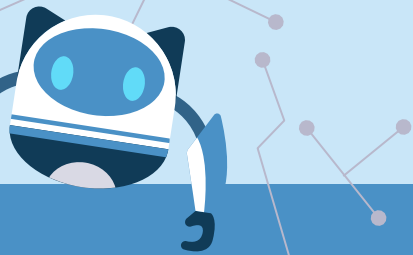
- Holds map of all piece locations in play ✓
- Can respond to a request of a specified piece location (Scrapped)
- Validates if a given move is valid ✓
- Track locations of taken pieces ✓

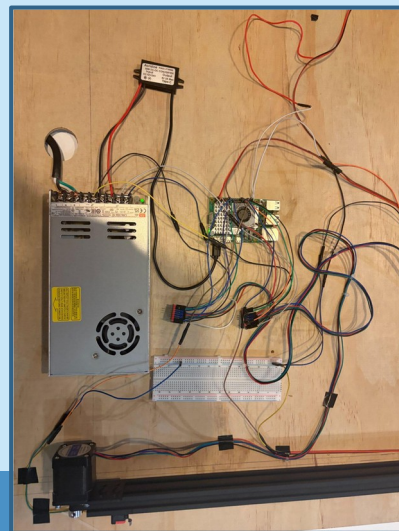
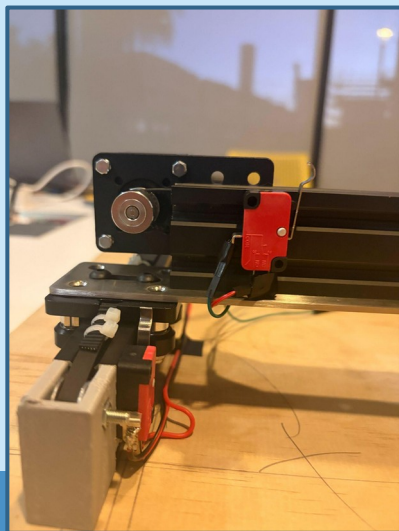
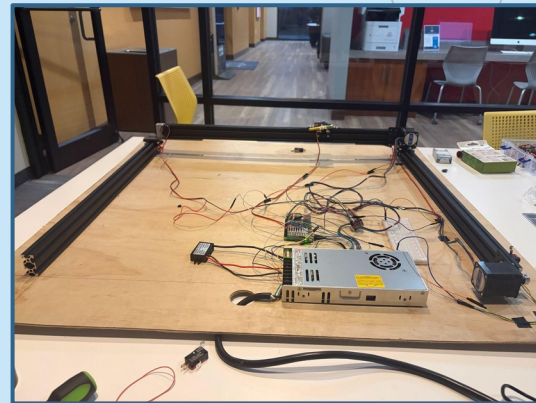
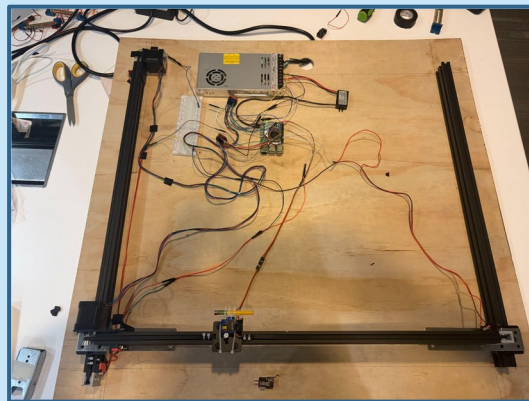
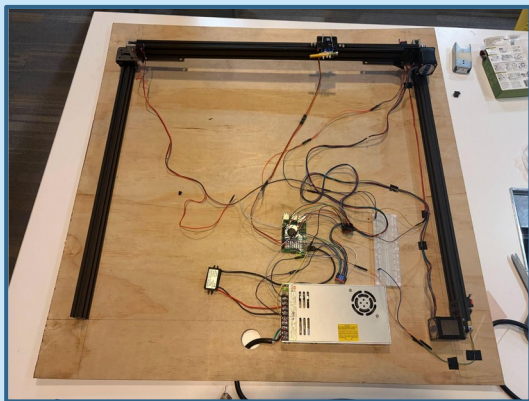
- Navigator Subsystem:

- Take input from planning ✓
- Plan the path for EM gantry arm
- Publish all new piece locations to Cartographer topic ✓
- Sends path to Motor Schema ✓

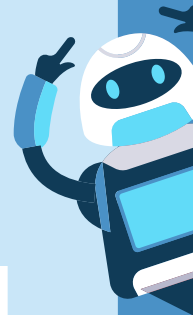
- Motor Schema:

- Converts path from Nav to motor output
- Can receive interrupt ✓
- Publishes completion message when done(Scrapped)





Challenges - how did we overcome them?



01

Speech-to-text

Been having issues setting up stt with permissions in the docker container

Docker container with admin privileges on pi found microphone

02

Python Env. Issues

Having issues with the python environment in ROS2 workspace

Removed all venvs and globally installed any packages in docker

03

Woodworking

The team has no experience with wood working

We (Kevin) made it work!

04

Wiring

No Electrical Majors

There are magnitudes of resources online!

Next Steps

01

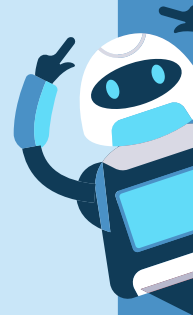
**Finish
navigation with
other pieces**

02

**Fully implement
cosmetic board
components**

03

**Play against
robot!**



Demo Video

